EXAM SHIELD

Artificial Intelligence and Machine Learning Question Bank

Learning paradigms, model evaluation, neural networks and responsible AI.

13 questions: 8 multiple choice, 3 fill in the gap and 2 written. Total 22 marks.
Edition: with answers (staff copy, ready to import into Exam Shield).

How this paper is laid out

Questions are numbered in one sequence. Multiple-choice options are lettered with a full stop; each multiple-choice question is worth one mark unless stated.

Fill-in-the-gap questions show each blank as a line; they earn one mark per blank, shown in square brackets, and are marked automatically.

Written questions show their marks in square brackets and are marked by an examiner.

To use this paper in Exam Shield, open an exam, choose Questions and then Import from PDF, and upload this file. Review the preview and import.

Section 1: Artificial Intelligence and Machine Learning

Learning paradigms, model evaluation, neural networks and responsible AI.

1. Training a model on labelled input-output examples is called:

A. Clustering
B. Unsupervised learning
C. Supervised learning
D. Reinforcement learning

Answer: C

2. Which of these is an unsupervised clustering algorithm?

A. Logistic regression
B. Linear regression
C. k-means
D. Naive Bayes

Answer: C

3. Which metric is the harmonic mean of precision and recall?

A. Accuracy
B. Mean squared error
C. ROC AUC
D. F1 score

Answer: D

4. Which technique reduces overfitting by randomly switching off neurons during training?

A. Padding
B. Momentum
C. Pooling
D. Dropout

Answer: D

5. Which neural network architecture underlies modern large language models?

A. Transformer
B. Single perceptron
C. Convolutional neural network
D. Hopfield network

Answer: A

6. In reinforcement learning, what does an agent try to maximise?

A. Number of labels
B. Training loss
C. Learning rate
D. Cumulative reward

Answer: D

7. Splitting data into k parts and training k times, holding out a different part each time, is called:

A. k-fold cross-validation
B. Grid search
C. Bagging
D. Bootstrapping

Answer: A

8. Which activation function outputs max(0, x)?

A. Tanh
B. Softmax
C. ReLU
D. Sigmoid

Answer: C

9. Neural networks compute gradients by applying the chain rule backwards through the layers, an algorithm called _____. [1 mark]

Answer: backpropagation / back propagation / back-propagation

10. Data used to tune hyperparameters is the _____ set, and data kept aside for the final estimate of performance is the _____ set. [2 marks]

Answer: validation | test

11. A _____ matrix tabulates true positives, false positives, true negatives and false negatives. [1 mark]

Answer: confusion

12. Explain the bias-variance trade-off and how it relates to underfitting and overfitting. [5 marks]

13. Discuss two sources of unfair bias in machine-learning systems and how responsible-AI practice reduces them. [5 marks]